

Lab: The Binary Bracelet

Goal

To write a message in Binary Code using beads.

What You Need

- String or Pipe Cleaners.
- Beads of two colors (e.g., Red and Blue).
- A Binary Key (Chart).

Steps

1. The Key

Use this simple code:

- A = Red-Blue-Red
- B = Red-Red-Blue
- C = Blue-Red-Red (*Real binary is 8 bits, this is a simplified 3-bit version for kids!*).

2. The Message

Choose a word. (e.g., "DAD").

- D = Blue-Red-Blue
- A = Red-Blue-Red
- D = Blue-Red-Blue

3. The Build

String the beads in order. Tie it around your wrist. You are wearing secret robot data!

My Bracelet Log

Letter	Color Code
First Letter	Red-Blue-Red (A)
Second Letter	Blue-Blue-Red (T)

What We Learned

Computers store everything (pictures, games, texts) as long strings of two things (On/Off, Red/Blue). We call these **Bits**.